



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SWAGAT COCO NAMAKKAL

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

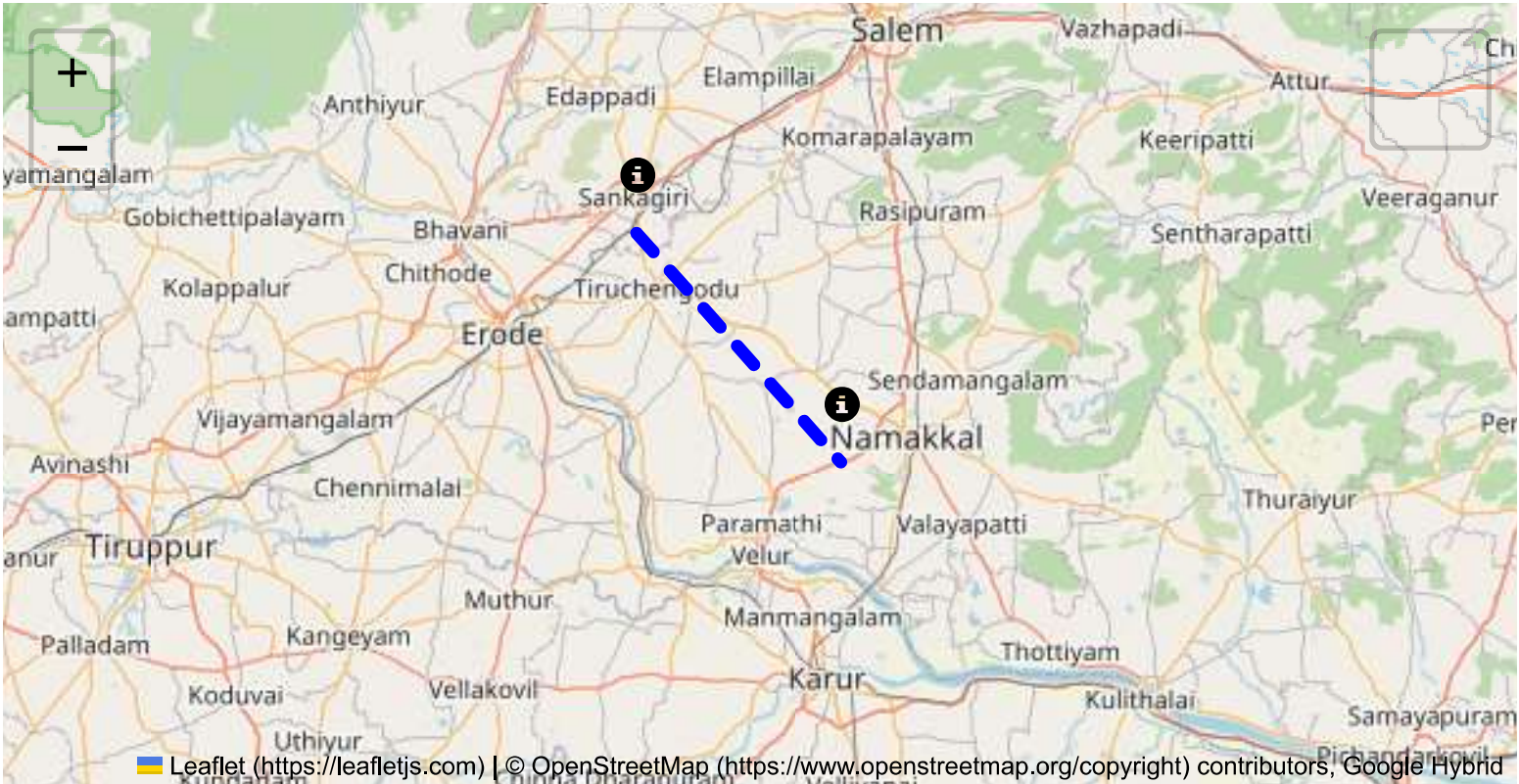
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

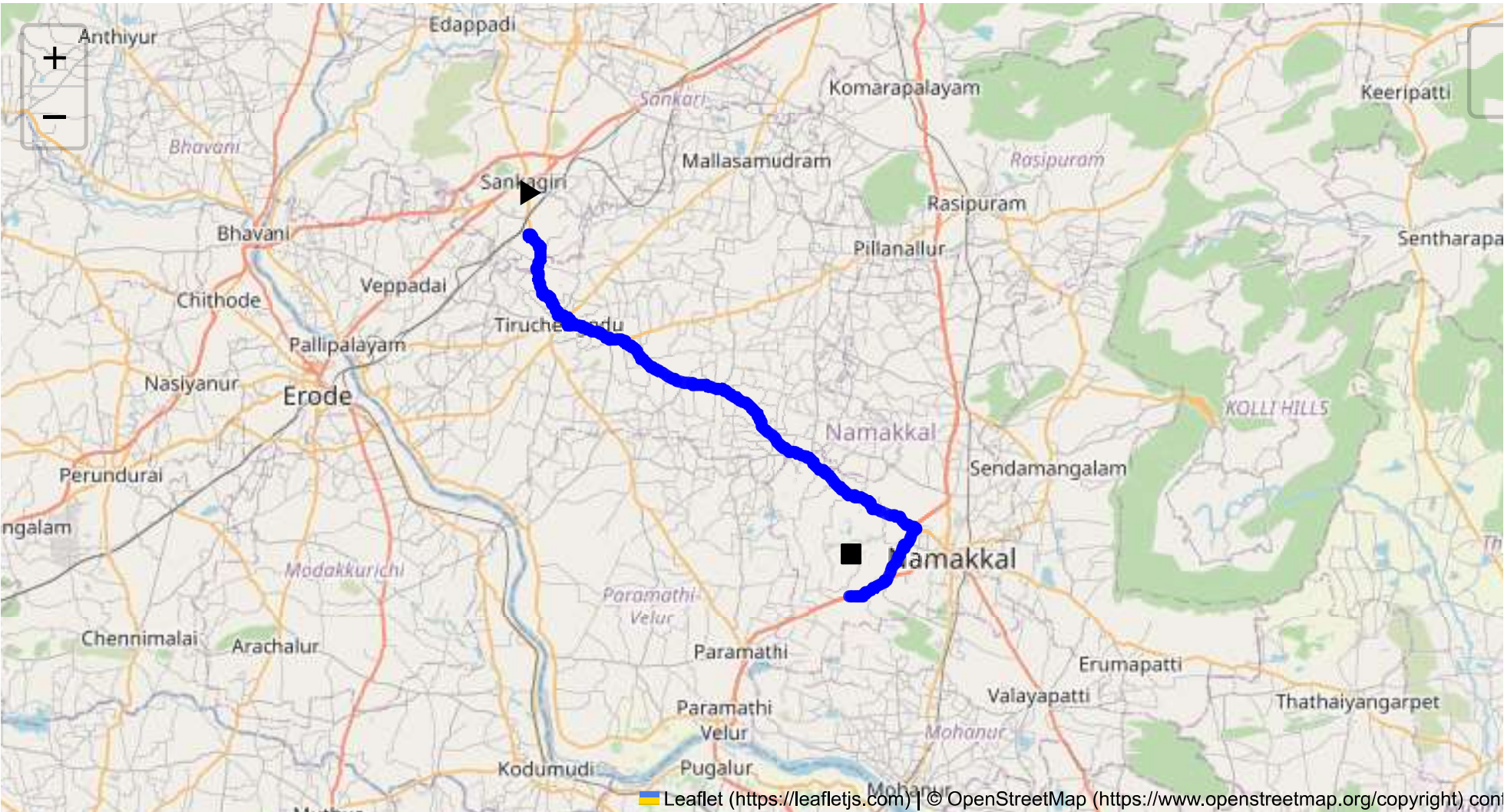
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 48.26 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.1919449, 78.096484)



Welcome to the Journey Risk Management Study

Route Analysis Report

1. Overview of the Route Map

The route from Sangagiri (CVQF+23W) to Thattankuttai (53VW+5JW) spans approximately 48.26 kilometers and typically involves traveling through rural and semi-urban areas in Tamil Nadu. The journey mainly

comprises National and State Highways, possibly including State Highway 86 (SH-86) and other connecting roads that lead towards Thattankuttai.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns:** Tamil Nadu experiences a tropical climate, with the southwest monsoon from June to September and the northeast monsoon from October to December.
- **Hazards:** Heavy rainfall during monsoons could lead to flooding and slippery road conditions. High temperatures and dry conditions can occur during the summer, from March to May.

3. Analysis of Traffic Patterns

- **Peak Hours:** Early morning (7:00 AM - 9:00 AM) and late evening (5:00 PM - 8:00 PM) typically see more traffic due to local commuting.
- **Congestion-Prone Areas:** Town centers and market areas along the route might experience congestion, particularly in Sangagiri and any urban settlements en route.

4. Assessment of Road Quality and Infrastructure

- **General Condition:** Roads in Tamil Nadu can vary from well-maintained highways to lesser-quality roads in rural areas. Potholes and narrow lanes are common in lesser-traveled sections.
- **Infrastructure:** Watch for narrow bridges and poorly marked intersections.

5. Suggestions for Alternative Routes

- In emergencies, consider using local roads that run parallel to major highways. Roads leading south towards Erode may offer alternative paths before veering back towards Thattankuttai.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Restrictions:** Transport of hazardous materials typically requires special permits in India. Routes through dense population centers often have time-bound restrictions.
- **Regulations:** Drivers must comply with the Central Motor Vehicle Rules, which include requirements for signage and safety equipment.

7. Overview of Historical Incidents

- No specific data on incidents involving heavy vehicles or hazardous materials is provided. However, it's crucial to consider general trends of vehicle accidents in the area, particularly at key intersections and rural roads.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Watch for protected agricultural lands, water bodies, and wildlife crossings. Ensure proper disposal and containment of hazardous materials to prevent environmental contamination.

9. Analysis of Communication Coverage

- **Potential Dead Zones:** Rural sections may experience limited mobile coverage. Ensure truck drivers have emergency communication devices covering multiple network providers.

10. Estimated Emergency Response Times

- **Urban Areas:** Emergency responses in areas like Sangagiri should be relatively swift, approximately 30-45 minutes.
- **Rural Segments:** Response times could increase to over an hour, especially in remote segments of the journey where medical and rescue facilities are sparse.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Thattankuttai features typical challenges associated with transporting hazardous materials over diverse road conditions. The primary risks include weather-related hazards, traffic congestion, and communication dead zones. By assessing alternative routes and preparing for environmental sensitivity, drivers can mitigate potential hazards. Compliance with local regulations governing hazardous material transport is crucial for safety along this route. Regular updates on traffic patterns and weather forecasts are advised to ensure uninterrupted and safe passage for heavy vehicles.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38190, 77.90254	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Blind Spot	Blind Spot	11.23936, 78.14267	10 KM/Hr
10	Turn	High	11.22817, 78.13576	15 KM/Hr
11	Turn	High	11.22822, 78.13561	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.37841, 77.90140



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.23936, 78.14267



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22817, 78.13576



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.22822, 78.13561

